

HW # 33

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CS-F

Prob: 01

$$3x_1 + 2x_2 - x_3 = 2$$

$$x_1 - 4x_2 + 3x_3 = -2$$

$$x_1 + 10x_2 - 7x_3 = 1$$

$$\begin{bmatrix} 3 & 2 & -1 \\ 1 & -4 & 3 \\ 1 & 10 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$A \quad \quad \quad x = b$

$$A^T A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & -4 & 10 \\ -1 & 3 & -7 \end{bmatrix} \begin{bmatrix} 3 & 2 & -1 \\ 1 & -4 & 3 \\ 1 & 10 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 12 & -7 \\ 12 & 120 & -84 \\ -7 & -84 & 59 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 3 & 1 & 1 \\ 2 & -4 & 10 \\ -1 & 3 & -7 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 22 \\ -15 \end{bmatrix}$$

$$(A^T A) \hat{x} = A^T b$$

$$\left[\begin{array}{ccc|c} 11 & 12 & -7 & 5 \\ 12 & 120 & -84 & 22 \\ -7 & -84 & 59 & -15 \end{array} \right] \sim \left[\begin{array}{ccc|c} 11 & 12 & -7 & 5 \\ 1 & 10 & -7 & 11/6 \\ -7 & -84 & 59 & -15 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 10 & -7 & 11/6 \\ 11 & 12 & -7 & 5 \\ -7 & -84 & 59 & -15 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 10 & -7 & 11/6 \\ 0 & -98 & 70 & -91/6 \\ 0 & -14 & 10 & -13/6 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 10 & -7 & 11/6 \\ 0 & 1 & -5/7 & 13/84 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 = -10x_2 + 7x_3 + \frac{11}{6}$$

$$\boxed{x_2 = \frac{5}{7}x_3 + \frac{13}{84}} \Rightarrow x_1 = -\frac{50}{7}x_3 + 7x_3 + \frac{11}{6} - \frac{130}{84}$$

$$x_1 = -\frac{1}{7}x_3 + \frac{11}{6} - \frac{130}{84}$$

$$\boxed{x_1 = -\frac{1}{7}x_3 + \frac{2}{7}}$$

$$\hat{x} = \begin{bmatrix} -1/7 x_3 + 2/7 \\ 5/7 x_3 + 13/84 \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} -1/7 \\ 5/7 \\ 1 \end{bmatrix} x_3 + \begin{bmatrix} 2/7 \\ 13/84 \\ 0 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 1/7 \\ 73/84 \\ 1 \end{bmatrix}$$

when $x_3 = 1$
∴ out of infinitely many

$$\begin{aligned} \hat{b} &= A \hat{x} \\ &= \begin{bmatrix} 3 & 2 & -1 \\ 1 & -4 & 3 \\ 2 & 10 & -7 \end{bmatrix} \begin{bmatrix} 1/7 \\ 73/84 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 7/6 \\ -1/3 \\ 11/6 \end{bmatrix} \end{aligned}$$

$$e = b - \hat{b} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 7/6 \\ -1/3 \\ 11/6 \end{bmatrix}$$

$$e = \begin{bmatrix} 5/6 \\ -5/3 \\ -5/6 \end{bmatrix}$$

$$\|e\| = 2.0412$$

Prob 2

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

a) $A_L^{-1} = (A^T A)^{-1} A^T$
 $= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ as calculated in previous how

b) $A_R^{-1} = A^T (A A^T)^{-1}$

$$A A^T = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 16 & 0 & 4 \\ 0 & 4 & 2 \\ 4 & 2 & 2 \end{bmatrix}$$

$(A A^T)^{-1}$ does not exist as $\det = 0$